PilotFish Call Route Processor – Complete Reference (26R1.11 WAR)

Derived from decompiled eip.war.hs.26R1.11

August 4, 2026

At a glance (plain English)

This step **invokes another route** by triggering a named listener – either synchronously (waiting for a response) or asynchronously (fire-and-forget). It supports looping the callout based on a boolean condition.

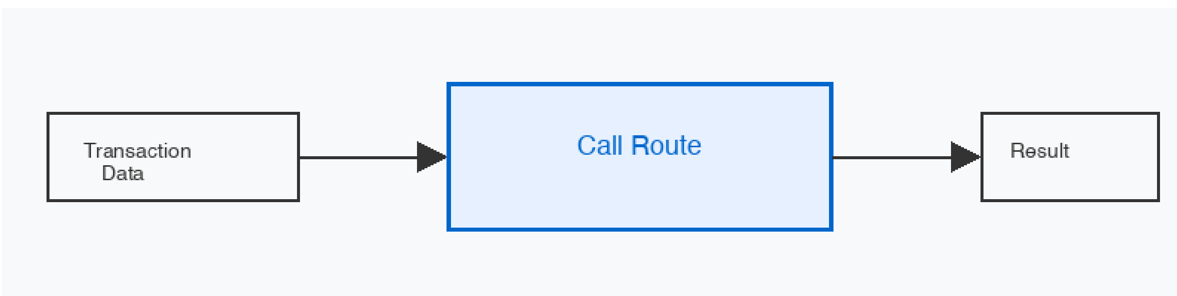

Figure 1: At a glance (plain English)

- **Triggers** a named Programmable/Triggerable listener to start another route.
- **Synchronous mode** waits for the response and replaces transaction data.
- **Asynchronous mode** fires and continues without waiting.
- **Looping** enables repeated synchronous callouts until a condition is met.

Purpose

Deep dive into **Call Route** (`CallRouteProcessor`) in **`eip.war.hs.26R1.11`**.

Provenance	Value
WAR	`eip.war.hs.26R1.11`
Module JAR	`xcs-modules-internal-1.0.1.jar`
Decompiler	CFR 0.152
Implementation class	`com.pilotfish.eip.modules.internal.CallRou`
Processor type string	`Call Route`
Categories	`PILOT_FISH`

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1. High-level behavior (`processData`)

1. If **synchronous**:
   a. If **looping enabled**:
      - Read max iterations from config.
      - While loop condition is true AND iterations < max:
        - Call doCallout (triggerAndWait).
        - Increment iteration.
   b. Else: single doCallout.
   c. doCallout: TriggerableListenerUtils.triggerAndWait with timeout; null response = timeout EIPException.
2. If **asynchronous**:
   a. Deep-clone transaction data.
   b. Trigger listener with clone (fire-and-forget).
   c. Call data.newCallout() to track.
3. Return data (synchronous: response data; async: original data).

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2. Configuration reference

3.1 Listener Name

Property	Value
UI label	Listener Name
Tag	`Listener`
Type	Listener reference
Required	yes
Tooltip	Specifies the programmatically-invoked listener which will initialize the route.

3.2 Synchronous

Property	Value
UI label	Synchronous
Tag	`Synchronous`
Type	Boolean
Default	`true`
Tooltip	Specifies if this call-out is synchronous (generates a synchronous response)

3.3 Timeout (sec)

Property	Value
UI label	Timeout (sec)
Tag	`Timeout`
Type	Integer
Default	`60`
Min/Max	5 / Integer.MAX_VALUE
Visible when	Synchronous = true
Tooltip	Amount of time permitted before considered a failure. Does not abort.

3.4 Enable Looping

Property	Value
UI label	Enable Looping
Tag	`LoopCallout`
Type	Boolean
Default	`false`
Visible when	Synchronous = true

3.5 Loop Condition

Property	Value
UI label	Loop Condition
Tag	`LoopCondition`
Type	Boolean (expression)
Default	`true`
Visible when	Synchronous = true AND Enable Looping = true

3.6 Max Loops

Property	Value
UI label	Max Loops
Tag	`MaxIterations`
Type	Integer
Default	`5`
Visible when	Synchronous = true AND Enable Looping = true

3.7 Inherited AbstractProcessor options

Tag	Label	Type	Default
`ExecuteProcessor`	Conditional Execution	Boolean	`true`

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3. Transaction attributes

None written directly. Called route may modify attributes on the shared transaction.

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4. Known quirks and caveats

1. **Timeout does not abort** – the called route keeps running even after timeout.
2. **Null response = timeout** – if triggerAndWait returns null, EIPException is thrown.
3. **Async clones data** – deep clone may be expensive for large payloads.
4. **Loop condition evaluated each iteration** – can reference attributes set by called route.
5. **Listener must be triggerable** – must be Programmable Trigger or similar.

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5. Operational recommendations

Goal	Guidance
Sub-route pattern	Use synchronous mode for request-response sub-flows
Background processing	Use async for non-blocking parallel work
Polling loops	Use looping with a condition that checks response attributes
Timeout tuning	Set timeout higher than expected sub-route duration

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6. Source index (this WAR)

Topic	Path
Call Route	`com/pilotfish/eip/modules/internal/CallRouteProces`
Module JAR	`xcs-modules-internal-1.0.1.jar`

Deep-dive documentation – Call Route Processor – 26R1.11 – August 2026.